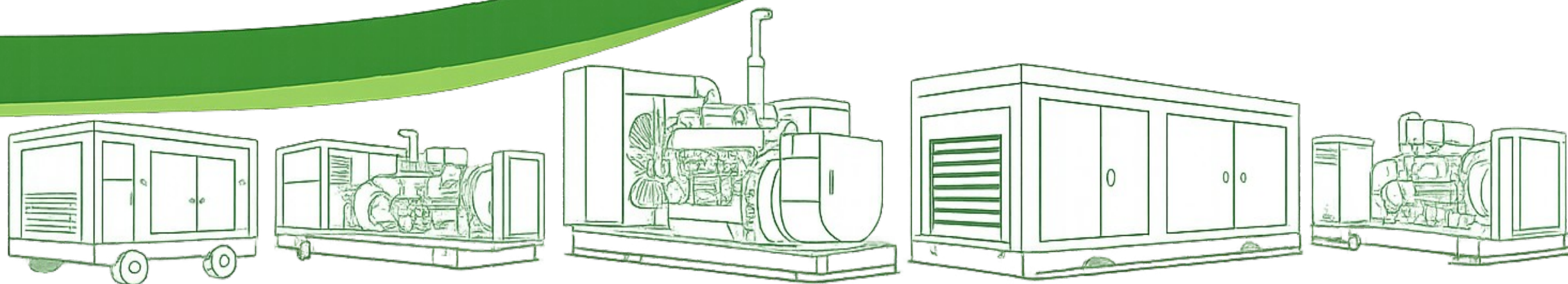


REDUCING GHG EMISSIONS FROM DIESEL GENERATORS USING GreenX™ HYDROGEN FUEL ASSIST RETROFIT

DECARBONISATION WITH PROFIT

A practical, low-capex pathway to reduce diesel consumption,
Exhaust emissions and tCO₂e for DGs from 20 to 4000 kVA



The Problem: DGs Under Severe Emission Pressure

Key Challenges

Tightening emission norms

Stricter CPCB, EPA, and global regulatory mandates now require rapid reduction of PM, smoke, NOx, and overall GHG emissions, forcing industries to act urgently.

Ineffective Post-Emission Solutions

Available retrofits (RECD) are **expensive**, have **high maintenance needs**, and often **increase fuel consumption**, adding to tCO₂e instead of reducing it.

No Payback on Compliance Investments

Most post-treatment systems only suppress visible smoke but provide **no fuel savings and no financial return**. Investments are effectively **sunk costs**.

Continuous DG Usage = Continuous Pollution

Critical sites rely on DGs for backup or continuous power, causing **constant emissions**, especially from **older 10–20-year generators** that remain in operation with no practical path to decarbonisation.

Industrial DG Reality

Regulatory Risk:

High (non-compliance penalties rising)

Cost of Post-Treatment Retrofits:

₹8–50 lakh per DG set (typical)

Fuel Impact of SCR/Filters:

1–7% increase → Higher CO₂ + tCO₂e

Payback:

None (compliance-only expense)

Introducing GreenX™ Hydrogen Fuel Assist Retrofit

Key System Features

- ✓ **Safe, On-Demand Hydrogen Generation**
Hydrogen produced as needed, eliminating risks of storing large quantities of flammable gas by using its only consumable - Water
- ✂ **Rapid Installation**
System can be installed in less than 4 hours, minimizing downtime
- 📈 **High Technology Readiness Level**
Proven technology with TRL 8-9, extensively tested across various industrial applications
- ♻ **Broad Applicability**
Compatible with DG sets ranging from 20 to 4000 kVA of any make and any age.



System Benefits

- 🛢 3-15% reduction in diesel consumption based on load
- 🌿 Significant reduction in PM, NO₂, CO, and HC emissions
- ⚙ Optimizes combustion without engine modification
- 🛡 No need for high-pressure hydrogen storage – Inherently safe

How GreenX™ Works



Water Electrolysis

System splits water (H_2O) into hydrogen (H_2) and oxygen (O_2) gas through electrolysis



Gas Infusion

Electronically metered
Generated gas mixture
blended into diesel engine's
air intake manifold



Combustion Optimization

Metered Hydrogen improves
combustion by increasing
flame speed and reducing
ignition delay

Key Benefits

 **Lower Diesel Consumption** - Optimized combustion leads to more efficient fuel usage & corresponding tCO₂e reduction.

 **More Complete Burn** - Ensures thorough utilization of diesel fuel

 **Reduced Emissions** - Decreased PM, NO₂, CO, and HC emissions

 **Stable Power Generation** - Particularly under varying load conditions

Smart Control & Safety Architecture



Adaptive Hydrogen Dosing

Intelligently adjusts hydrogen dosage based on real-time engine conditions for optimal performance and safety

No Hydrogen Storage

Hydrogen produced on-demand and immediately consumed, eliminating risks associated with high-pressure storage



Built-in Safety Logic

Engineered for continuous remote operation with comprehensive safety protocols for demanding environments



Integrated Monitoring

Seamlessly integrated with SGT GreenVision™ for real-time monitoring of fuel usage, load profiles, and emission signatures



Multiple Shutdown Protections

Includes safeguards against overheating, backflow, and electrical anomalies to ensure system integrity



Proven Technology

SGT's Smart Control Architecture has been successfully tested & deployed across various industrial applications, ensuring reliable operation in offshore, 24/7 environments

Why It Works Exceptionally Well on Old DGs

⚠ Challenges with Aging DGs

- 💧 **Poor Atomisation**
Diesel fuel not atomized as finely, leading to incomplete combustion
- 🕒 **Timing Drift**
Mechanical wear causes injection timing to drift, compromising combustion
- ☁ **Combustion Losses**
Incomplete burning results in wasted fuel and higher emissions
- 💧 **Soot Formation**
Consequence of inefficient combustion is soot and carbon deposits

The Outcome for Old Engines:

- ⚙ **Higher Efficiency**
- ☁ **Lower Soot Deposits**
- ⚡ **Better Power Stability**
- 🛢 **Reduced Fuel Wastage**

✅ How GreenX™ Compensates

- 🔥 **Accelerating Flame Propagation**
Hydrogen's high flame speed ensures rapid ignition and burn of diesel fuel
- 📦 **Near-Complete Diesel Burn**
Enhanced combustion allows thorough utilization of diesel fuel
- 📈 **Improved Efficiency**
Higher overall efficiency despite mechanical limitations
- 🛡 **Enhanced Stability**
Improved power stability and responsiveness under varying loads

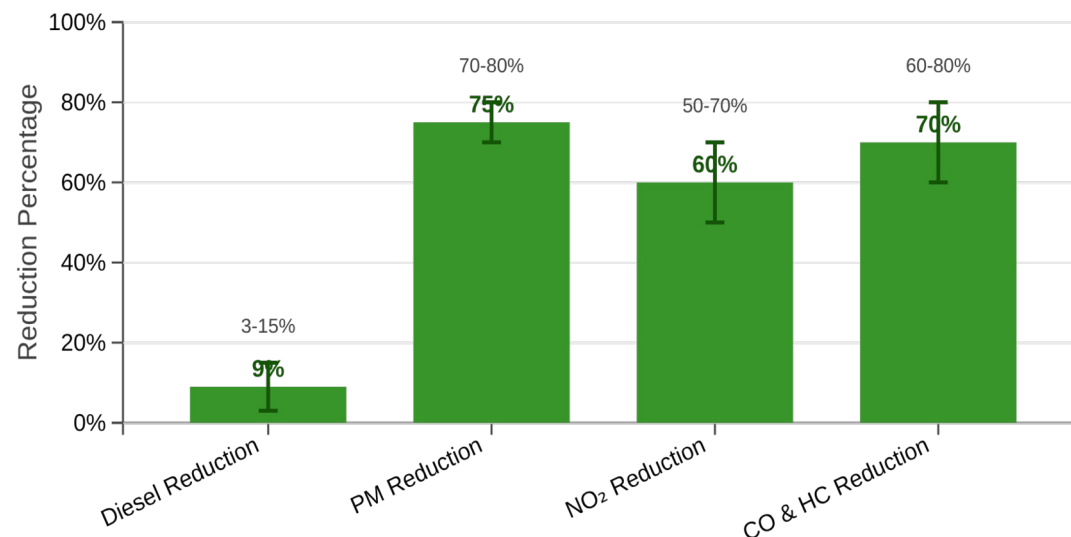
Proven Validation Across DG Sizes

Comprehensive Testing & Validation

-  **Range:** Tested on diesel generators from **125 kVA to 1500 kVA**
-  **Fuel Reduction:** Consistent **3% to 15%** reduction in diesel consumption
-  **Particulate Matter:** **70% to 80%** reduction in PM emissions
-  **NO₂ & Other Emissions:** **50% to 70%** reduction in NO₂ and **60% to 80%** reduction in CO & HC

Predictable Performance

GreenX™ demonstrates consistent and reliable performance across all DG sizes, making it a versatile solution for various decarbonization needs.



✔ **TRL 8-9** technology readiness level

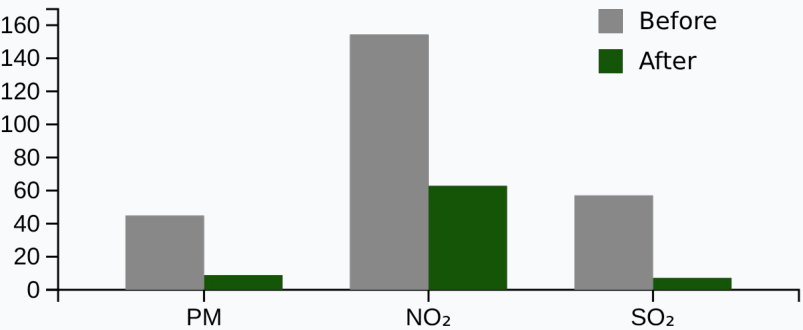
✔ **Rapid installation** in less than 8 hours

✔ Compatible with **20-4000 kVA** DG sets

Real Stack Test Results

⚡ 1500 kVA DG Results

Tested by NABL Accredited Labs



☁️ **PM Reduction:**
80%

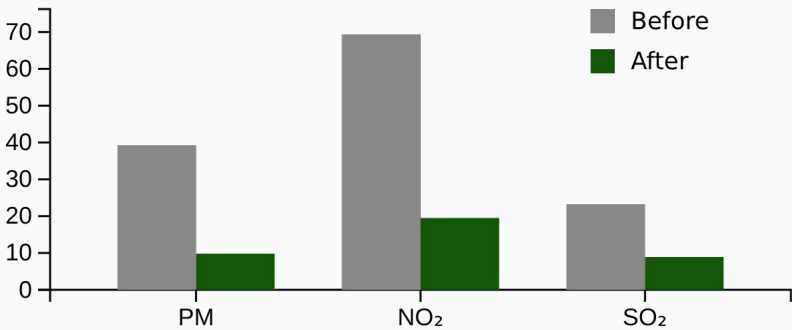
☁️ **NO₂ Reduction:**
59%

☁️ **SO₂ Reduction:**
87%

⛽ **Fuel Efficiency:**
8.5%

⚡ 380 kVA DG Results

Tested by NABL Accredited Labs



☁️ **PM Reduction:**
75%

☁️ **NO₂ Reduction:**
72%

☁️ **SO₂ Reduction:**
62%

⛽ **Fuel Saving:**
3-12% (based on load)

✓ **Combined Results: Significant Emission Reductions & Fuel Efficiency Gains**



8.5% Average Fuel Savings



Fuel Savings & Carbon Tax Benefits

Fuel Efficiency

Diesel Reduction **Estimated -**



8.5%

Typical range 3-15% based on load

tCO₂e Reduction **Estimated 8.5%+**



Environmental Impact



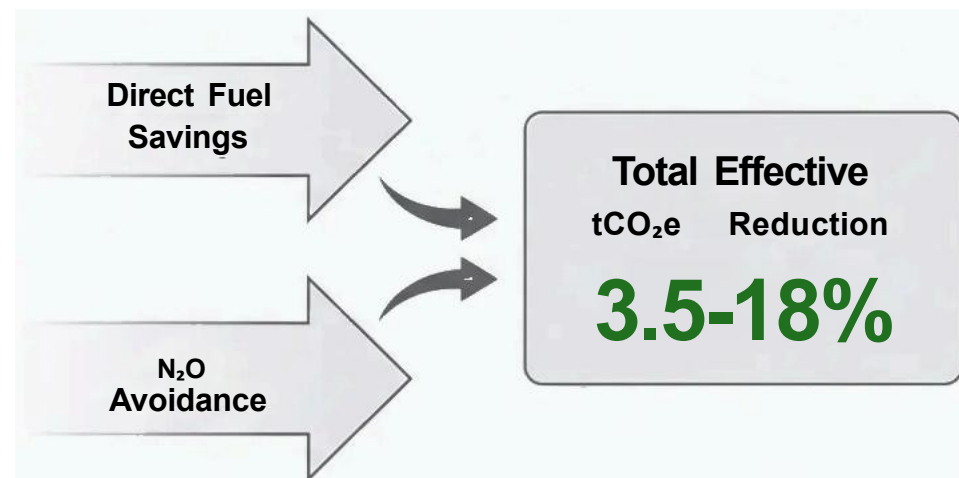
GreenX technology can help meet regulatory requirements and satisfy stakeholder demands for sustainability



Supports Scope 1 & 3 ESG emission reduction goals

External Retrofits using Oxidation Catalysts create N₂O, a harmful GHG that adds to tCO₂e and also use more Diesel fuel due to back pressure on engine.

With GreenX, no N₂O getting created and fuel saved - Combined tCO₂e +ve Impact



No Fuel Switching Required

Achieve meaningful carbon reductions while continuing to use conventional diesel fuels, avoiding costly alternative fuel infrastructure.

Measurable Climate Benefits

All emission reductions are quantifiable and reportable for regulatory compliance and ESG reporting requirements.

Proof of Concept & Implementation Path

Proof of Concept

Duration: 1 day test on one DG supported by stack emission report from qualified Lab.

Flexible Deployment Options

CapEx Purchase

Direct capital expenditure purchase of GreenX™ units with annual AMC

Decarbonisation-as-a-Service

Small upfront deposit with monthly subscription where Number of DG more than 10

POC Deliverables

- ✓ Verified fuel savings report
- ✓ Comprehensive stack-test emissions report
- ✓ tCO₂e reduction assessment
- ✓ Projected ROI for full deployment

Strategic Value for SGT GreenX Customers

Fastest Pathway

Immediate reduction in DG emissions

Low CapEx

Minimal upfront investment required

High Reliability

Proven technology with robust design

Measurable Impact

Verifiable CO₂ reduction data

Cost of GreenX System : Significantly less than any available RECD

Contact & Next Steps

Get in Touch

Partner with SGT to become a decarbonization leader. Our GreenX™ Hydrogen Fuel Assist Retrofit offers a practical, high-ROI solution for immediate emission reduction on any old or new DG set without requiring significant capital investment or operational disruption.

Take the first step with our Proof of Concept to validate performance on your existing diesel generators and develop a tailored implementation roadmap.



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